Debanuj Nayak

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Education

2022 - Present **Boston University**, *Ph.D., Computer Science*, **Advisor** : Sofya Raskhodnikova

2016 - 2020 **IIT Gandhinagar**, *B.Tech (Hons.) Computer Science & Engineering*, CPI: 9.56/10.0

Research Interests

My research interests lie broadly in **theoretical computer science**, particularly in **differential privacy**, and **property testing**. I am interested in designing algorithms that remain useful under constraints such as privacy, limited access to data, and adversarial manipulation, with applications to **trustworthy and responsible AI**.

Publications

- **Homomorphism Testing with Resilience to Online Manipulations**, *Esty Kelman, Uri Meir, **Debanuj Nayak**, Sofya Raskhodnikova.*, International Conference on Randomization and Computation, RANDOM 2026, [Link]
- **Differentially Private Multi-Sampling from Distributions**, *Albert Cheu, **Debanuj Nayak**.*, Algorithmic Learning Theory, ALT 2025, [Link]
- **On Fair Division with Binary Valuations Respecting Social Networks**, *Neeldhara Misra, **Debanuj Nayak**.*, International Conference on Algorithms and Discrete Applied Mathematics, CALDAM 2022, [Link]
- **Non-clairvoyant Scheduling of Coflows**, *Akhil Bhimaraju, **Debanuj Nayak**, Rahul Vaze.*, International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks, WiOPT 2020, [Link]
  Best Paper Award

Preprints

- **Differentially Private Modeling of Disease Transmission within Human Contact Networks**, *Shlomi Hod, **Debanuj Nayak**, Jason R. Gantenberg, Iden Kalemaj, Thomas A. Trikalinos, Adam Smith.*, arXiv preprint, 2026, [Link]

Research Projects

May 2026 - Present **Private Coresets for Linear Regression**
Max Planck Institute for Security and Privacy Bochum

- Study differentially private coresets for linear regression. Coresets are compact summaries that approximately preserve the regression cost for every query.
- Develop private algorithms and lower bounds for ℓ_2 regression, focusing on the tradeoff between privacy and additive approximation error.

Jan 2026 - Present **Private Tolerant Distribution Testing**
Boston University

- Study the sample complexity of tolerant distribution testing under differential privacy.
- Develop private tolerant testers for fundamental problems such as closeness and independence testing, and analyze the additional sample complexity required by privacy and tolerance.

May 2024 - May 2026 **Homomorphism Testing over Groups with Online Manipulations**
Boston University

- Study property testing of group homomorphisms in the presence of an online adversary that can manipulate the function during the testing process.
- Design resilient testers with query complexities comparable to their non-adversarial counterparts, including group-specific bounds for several families of groups.

Jan 2023 - May 2026 **Differentially Private Modeling of Disease Transmission within Human Contact Networks**
Boston University

- Develop a privacy-preserving pipeline for modeling disease transmission on sensitive human contact networks by combining node differential privacy with exponential random graph models (ERGMs).
- Empirically evaluate the effect of privacy on downstream disease-spread simulations, showing that bias due to privacy is smaller than model misspecification bias and additional variance due to privacy can be small relative to sampling and simulation variance.

Sep 2022 - Dec 2024 **Differentially Private Multi-Sampling from Distributions**
Boston University

- Study the problem of generating multiple samples from an unknown distribution while protecting the privacy of the samples used as input.
- Develop algorithms and sample-complexity bounds for privately generating multiple outputs whose joint distribution approximates independent samples from the original distribution.

Professional Experience

Jul 2020 - June 2022 **Goldman Sachs - Analyst (Software Engineer)**, Developed enterprise applications and backend systems for internal business users using Java, SQL, React, Linux, and related technologies.

Teaching Experience

- CS237 Probability in Computing, Boston University, Spring 2023, Fall 2026
- CS235 Algebraic Algorithms, Boston University, Fall 2026
- CS535 Complexity Theory, Boston University, Spring 2026
- ES112 Computing with Python, IIT Gandhinagar

Achievements

- **KVPY Fellowship Award** - Achieved All India Rank 304 and selected for the Fellowship in 2015.
- **Dean's List IIT Gandhinagar** - Selected for the Prestigious Dean's List for academically meritorious students in Semesters I , II , III , IV and V.
- **IIT JEE Advanced 2016** - Secured All India Rank **AIR 2487** among 200,000 candidates.

Skills

- **Programming:** Python, Java, SQL
- **Languages:** English, Hindi, Bengali